CELL ORGANELLES

LEARNING OBJECTIVES

At the end of lecture student must be able to:

- Define Organelles.
- Enumerate Cell Organelles.
- Discuss Detailed Structure, Composition & Functions Of Each Cell Organelles (Mitochondria, Ribosome, Endoplasmic Reticulum, Golgi Complex & Lysosomes)
- Explain The Structure & Composition Of Cytoplasm & Cytosol.

DEFINITION OF CELL ORGANELLES

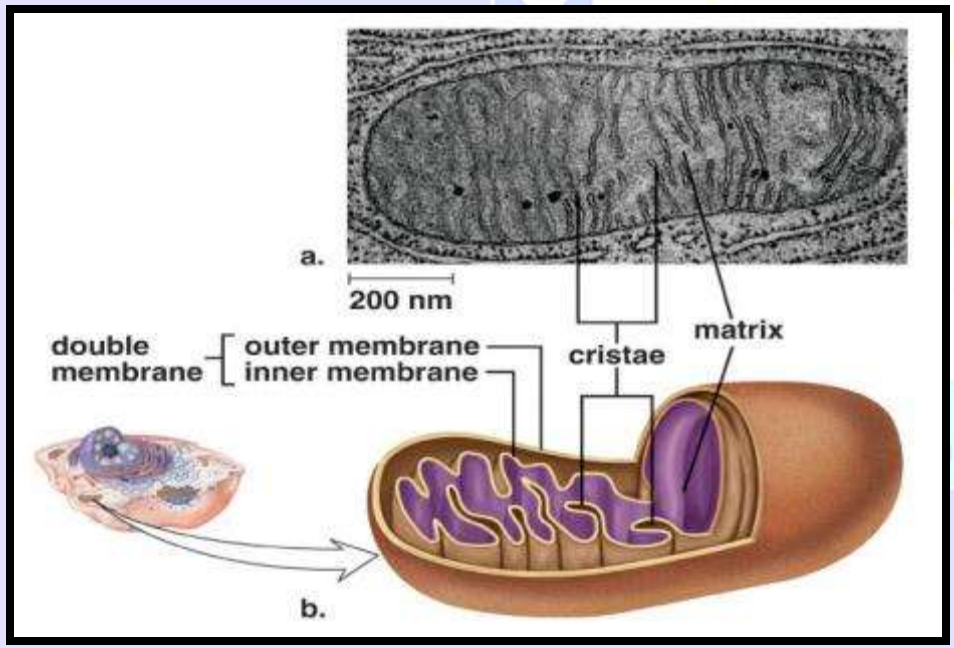
- Cell Organelle is a specialized subunit within a cell that has a specific function, and is usually separately enclosed within its own lipid bilayer.
- Organelles are identified by microscopy.
- There are many types of organelles, particularly in eukaryotic cells.

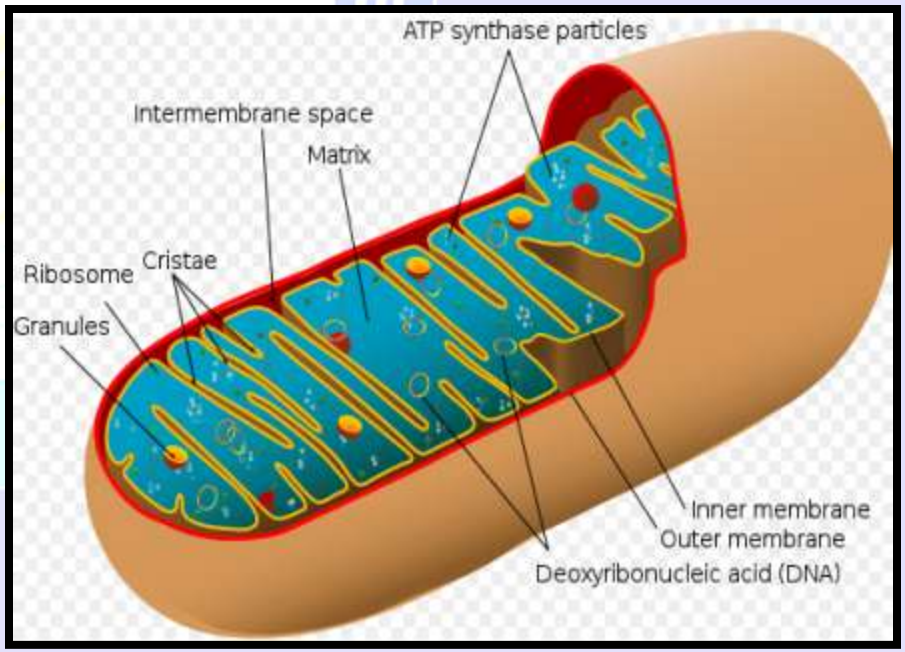
CYTOPLASMIC ORGANELLES

- Mitochondrion
- Ribosomes
- Endoplasmic reticulum (ER)
- Golgi apparatus
- Lysosomes
- Peroxisomes

MITOCHONDRIA

- ***Mitochondria*** are found in plant and animal cells.
- Mitochondria are bounded by a double membrane surrounding fluid-filled matrix.

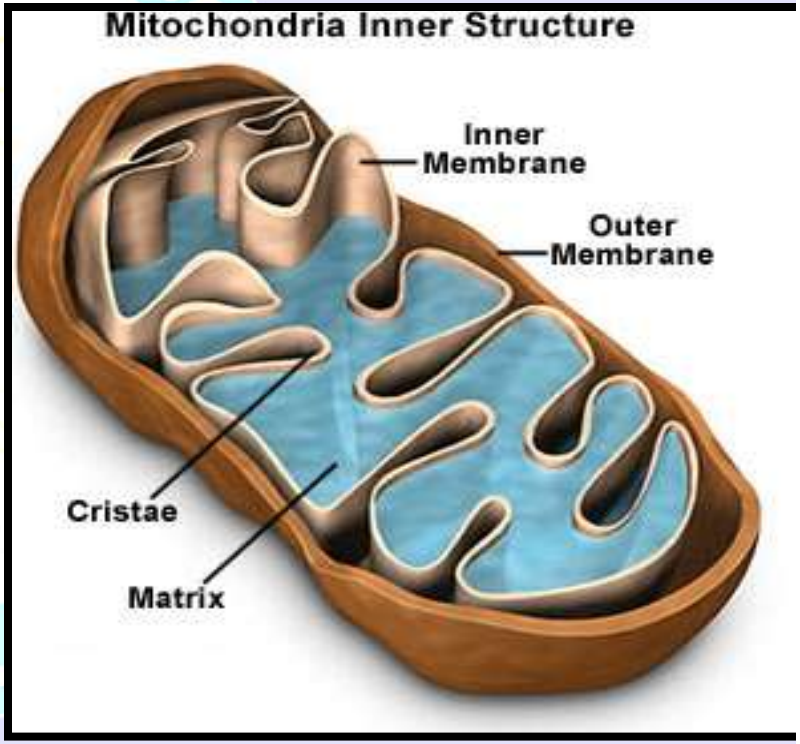




MITOCHONDRIA

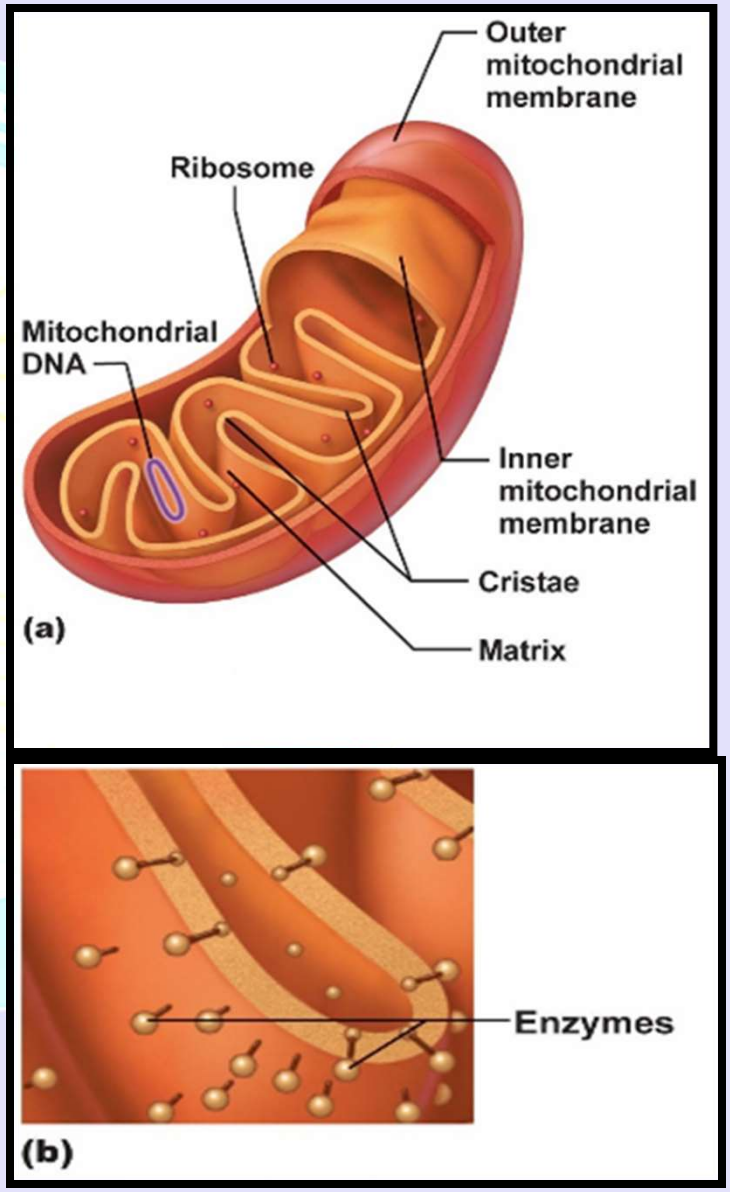
Mitochondria have:

- An **outer membrane** that encloses
- The entire structure
  - an **inner membrane** that encloses a fluid-filled **matrix**
  - between the two is the
  - **intermembrane space**
  - the inner membrane is elaborately folded with shelf like **cristae** projecting into the matrix.



MITOCHONDRION

- Generates most ATP
  - "power plant of the cell"
- Double membrane
  - outer smooth, sausage-shaped
  - inner folded with enzymes
- Contains own DNA, probably evolved from symbiotic bacteria



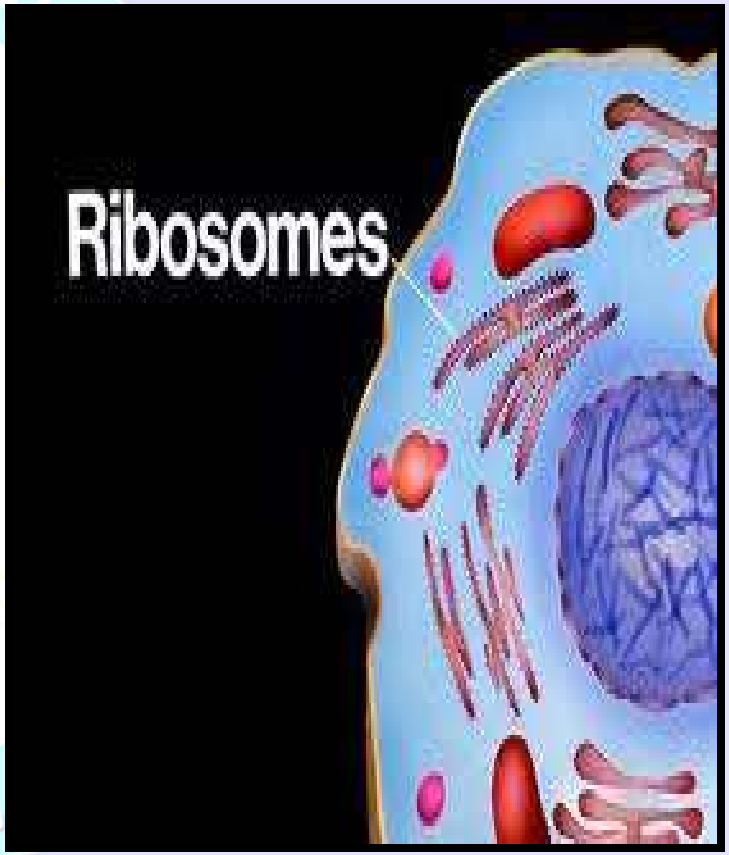
FUNCTIONS OF MITOCHONDRIUM.

The most prominent roles of mitochondrium is:

- to produce ATP (i.e., phosphorylation of ADP) through respiration.
- Regulation of the membrane potential
- Apoptosis-programmed cell death
- Calcium signaling (including calcium-evoked apoptosis)
- Cellular proliferation regulation
- Steroid synthesis.
- Certain heme synthesis reactions

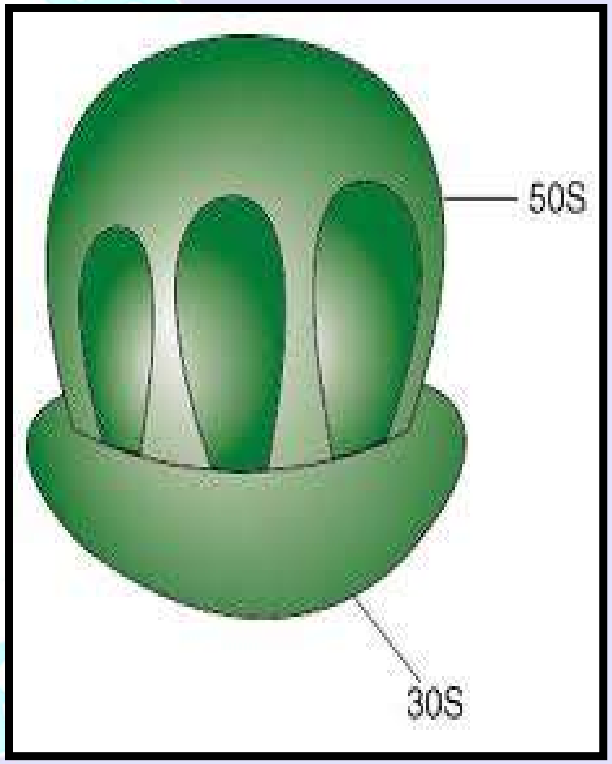
RIBOSOME

- Ribosome basically a protein factory.
- Subunits each have role in making of proteins
- To understand exactly what each subunit does, it's necessary to walk through protein synthesis step by step



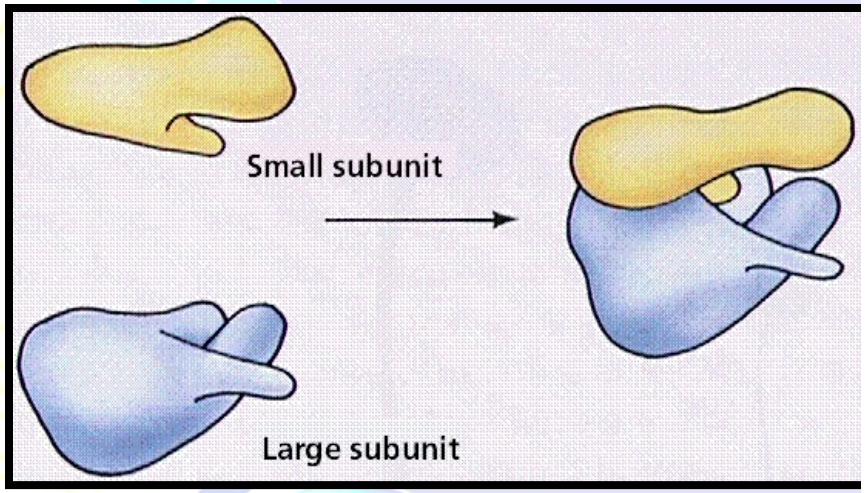
RIBOSOME

- Structures of the two ribosome subunits
  - The larger subunit
  - The smaller subunit
  - RNA's relation to their structure



RIBOSOME

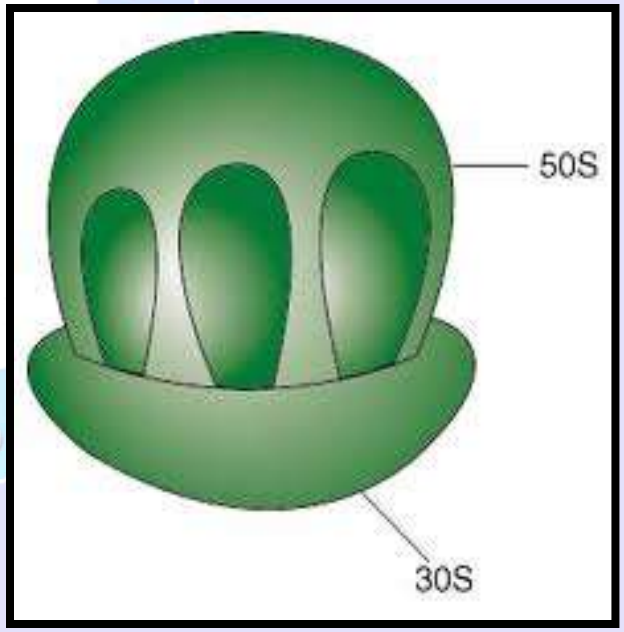
- Ribosome - protein synthesizer consisting of two subunits
- Larger one, “is 50S”,
- Smaller is “30S”
- (They look the same size here because of space restrictions.)



50S AND 30S???

- Related to their respective sizes. Numbers actually measures of how quickly each subunit sinks to the bottom of a container of liquid when spun in a centrifuge

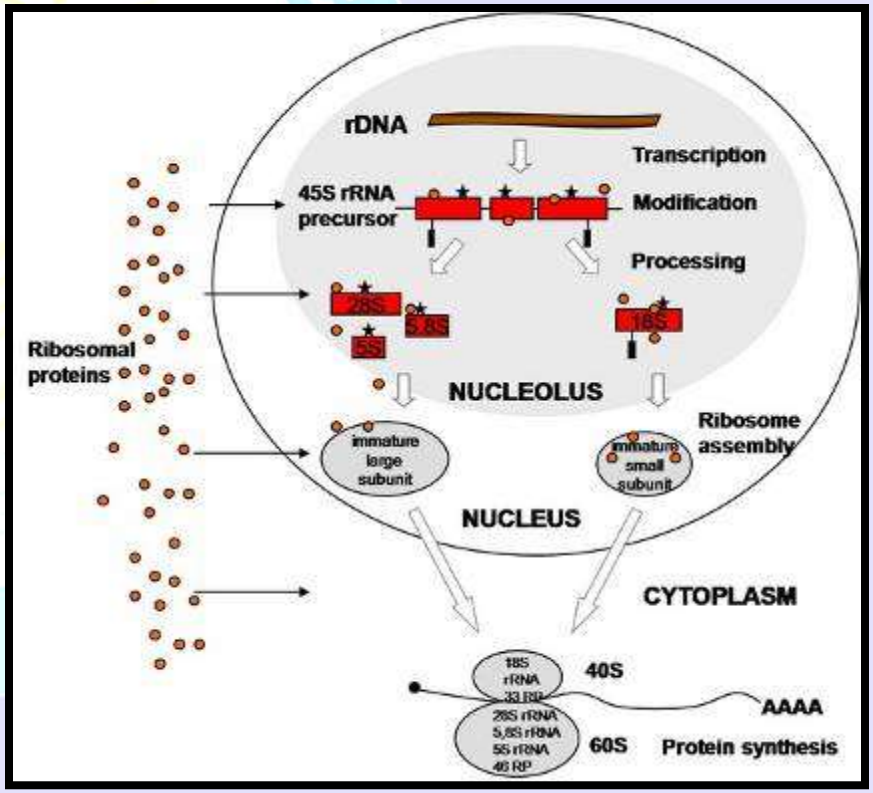
- One subunit smaller than other, but both are larger than average protein



PURPOSE OF RIBOSOME

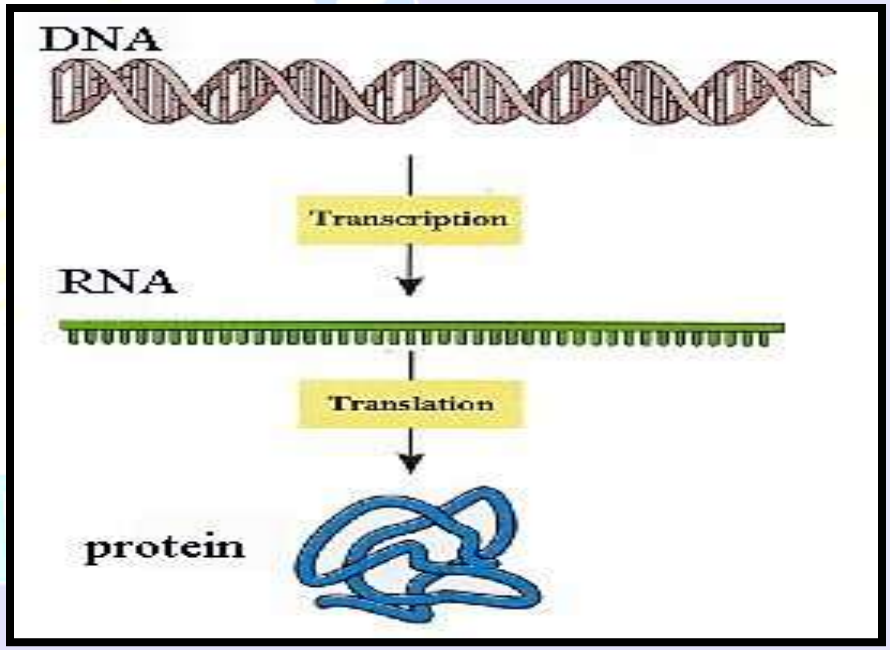
- *Ribosome basically a protein factory. Subunits each have role in making of proteins.*

- *To understand exactly what each subunit does, it's necessary to walk through protein synthesis step by step*



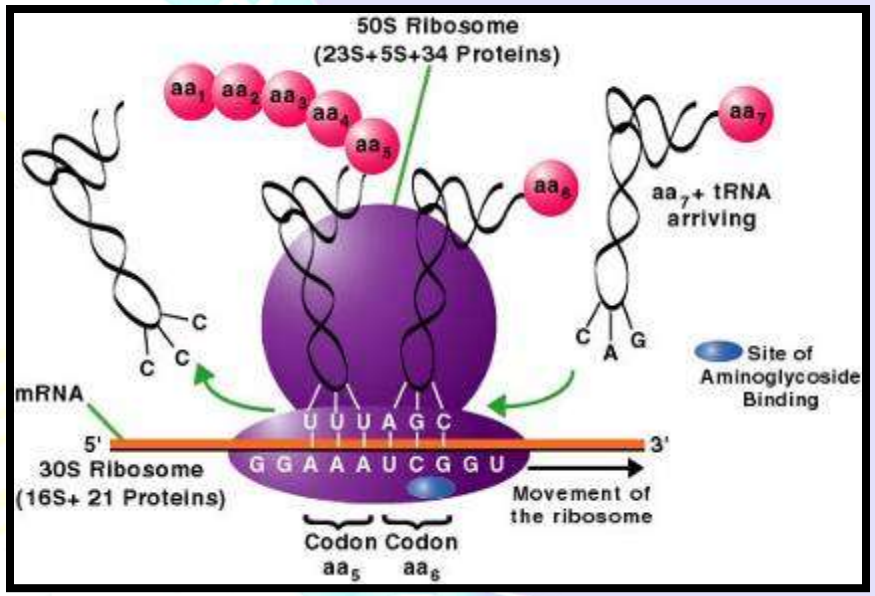
PROTEIN SYNTHESIS

- Process starts from DNA through “transcription”
- “Translation” is where ribosome comes in. Translation occurs when protein formed from code on mRNA
- Ribosome carries out the translation of the nucleotide triplets



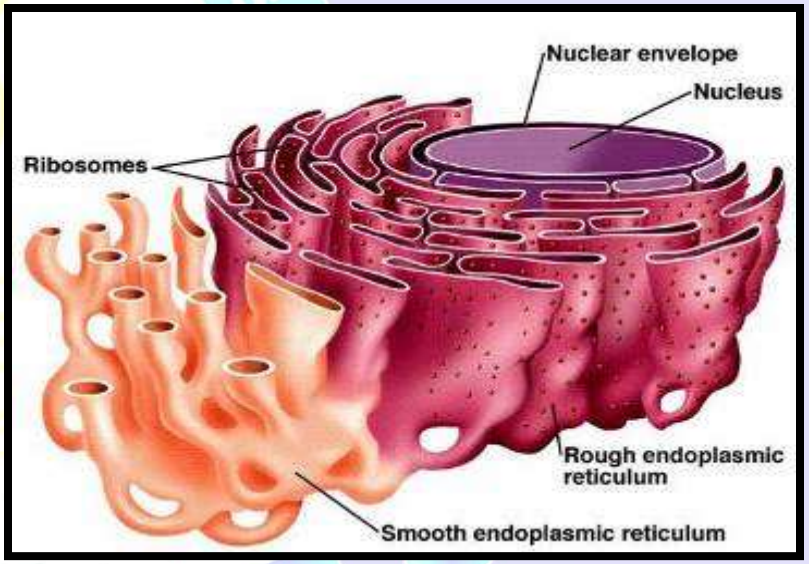
RIBOSOME AND RNA

- mRNA with code for proteins located at 30S subunit.
- tRNAs responsible for carrying amino acids to mRNA.
- Each tRNA has own nucleotide triplet which binds to matching triplet on mRNA,tRNA with code AAA (triple adenine) would match up with mRNA that has code UUU (triple uracil)



ENDOPLASMIC RETICULUM

- Endoplasmic reticulum (ER) forms an interconnected network of tubules, vesicles, and cisternae within cells.



  - **Rough Endoplasmic Reticulum:**
    - **Synthesize Proteins.**

-- Smooth Endoplasmic Reticulum:

- **Synthesize lipids and steroids.**

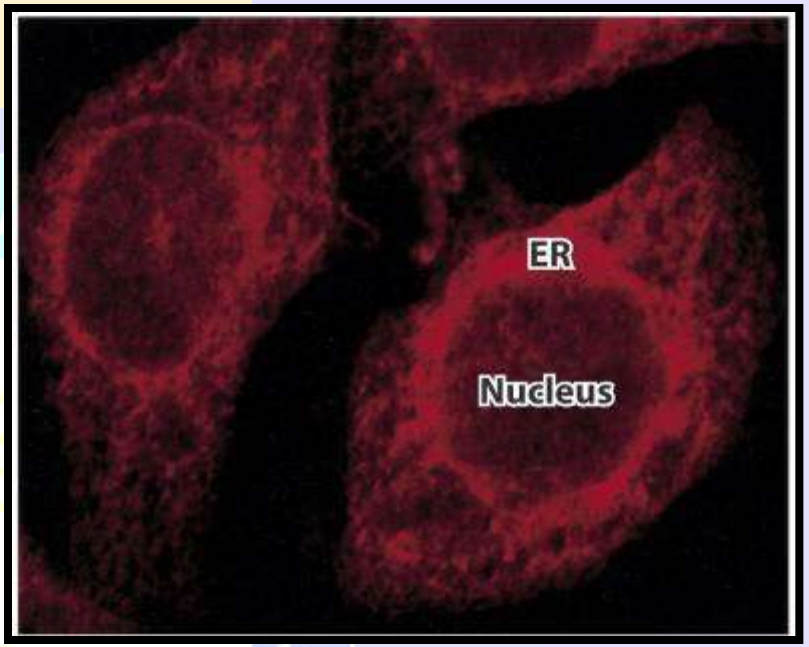
SMOOTH ENDOPLASMIC RETICULUM

FUNCTIONS:

- Synthesis of steroid hormones (endocrine cells of gonad and adrenal cortex)
- Detoxification of organic molecules (ethanol, barbiturates)
- Glucose release from liver
- Ca^{2+} sequestration



ROUGH ENDOPLASMIC RETICULUM



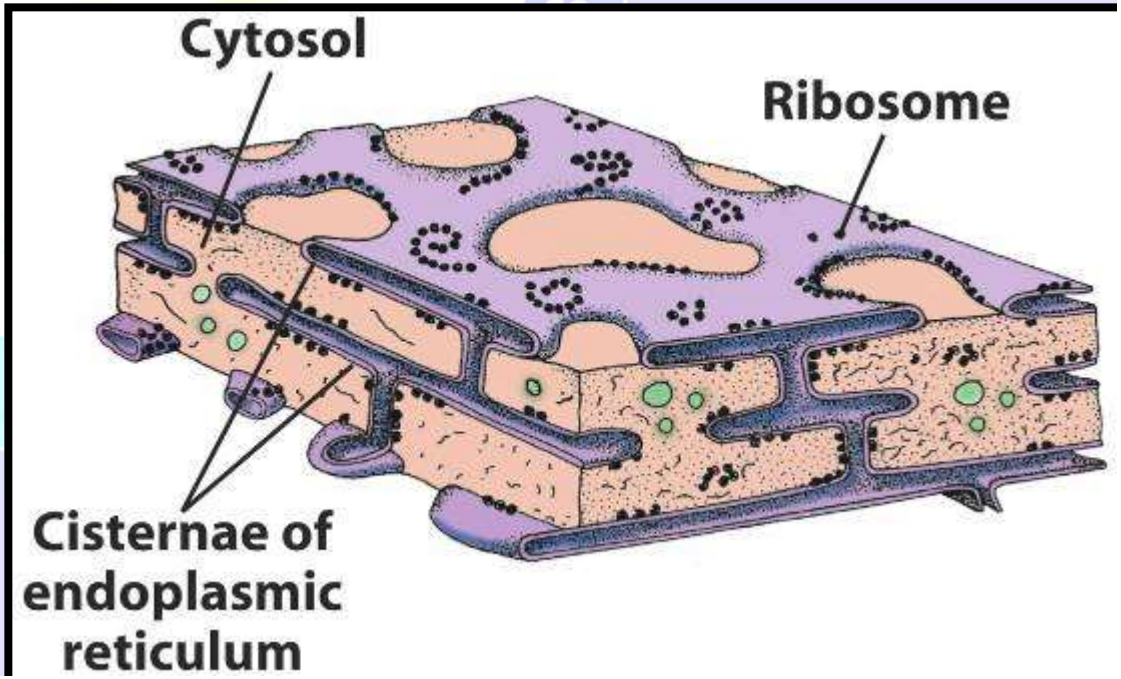
Functions:

starting point of BIOSYNTHETIC PATHWAY

Synthesis of:

- ***proteins***

PROTEIN MODIFICATION

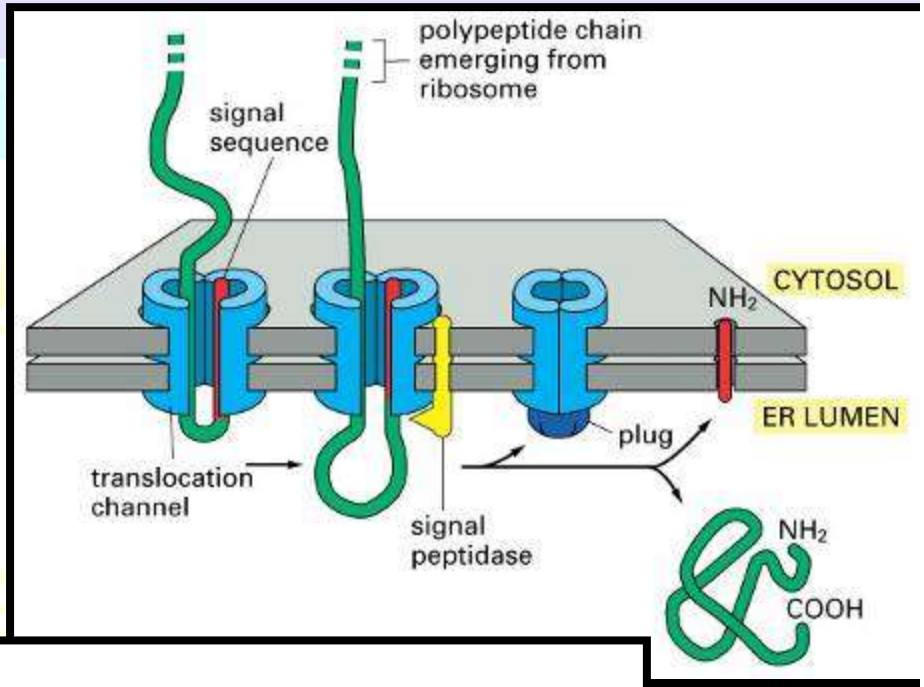


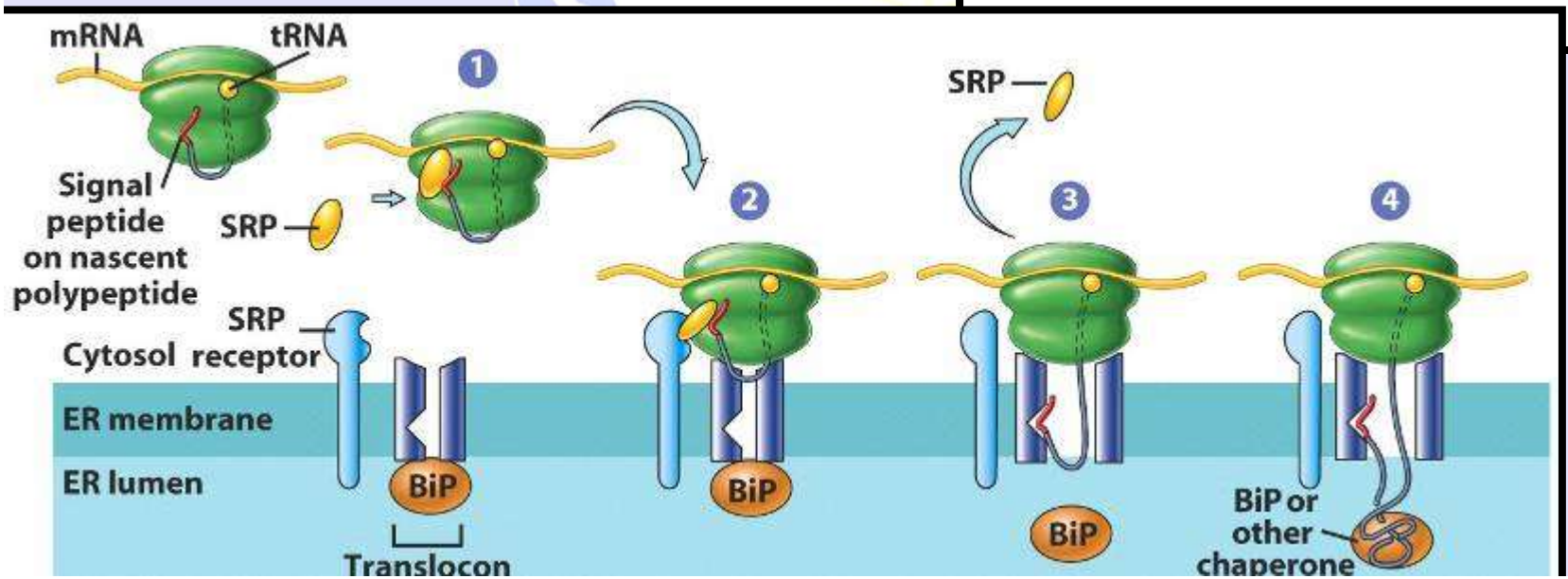
ROUGH ENDOPLASMIC RETICULUM

Protein synthesis on ribosomes attached to ER membrane:

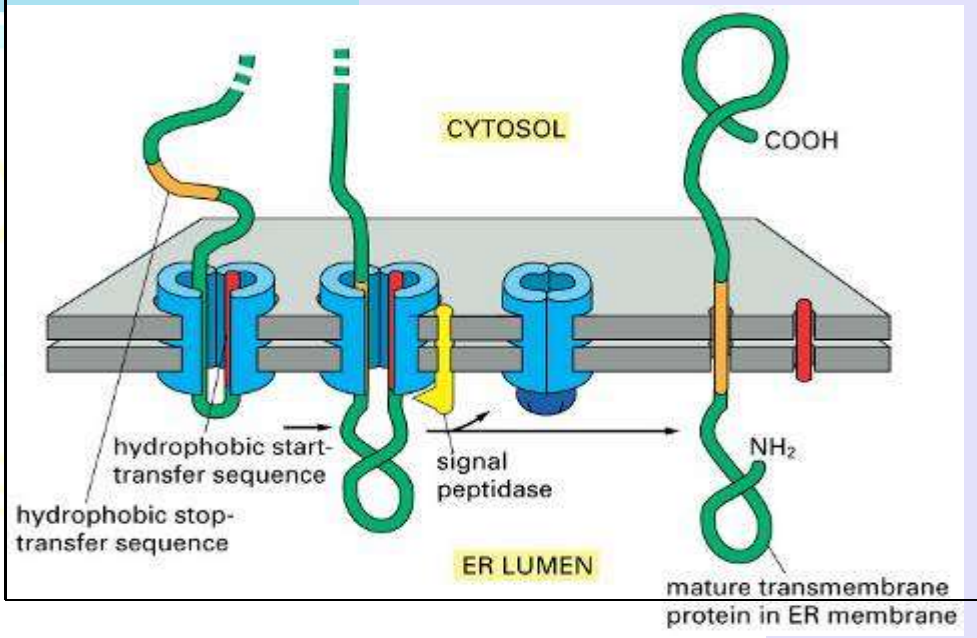
- Secreted proteins
- Integral membrane proteins
- Soluble proteins destined for life in a membrane bound compartment

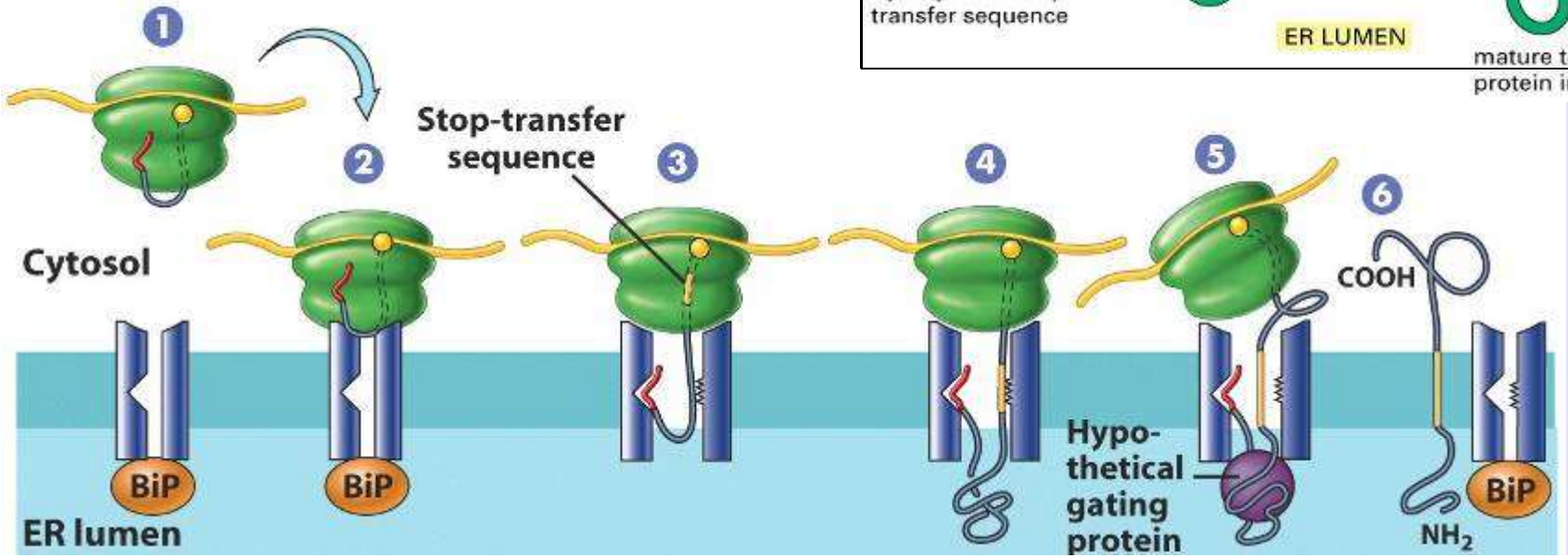
SYNTHESIS OF SECRETED PROTEIN ON ER MEMBRANE-BOUND RIBOSOME



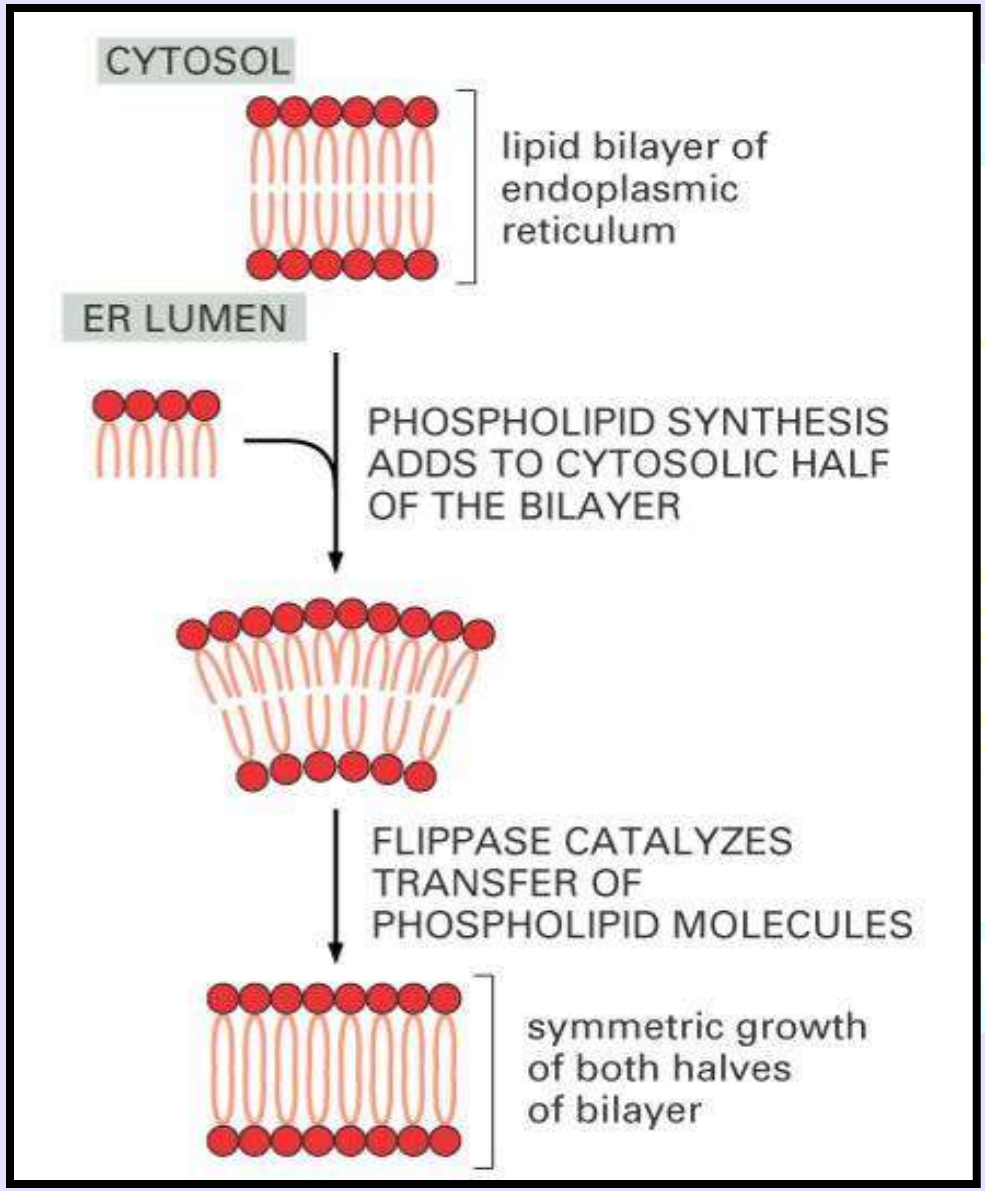


SYNTHESIS OF INTEGRAL MEMBRANE PROTEIN ON ER-BOUND RIBOSOME

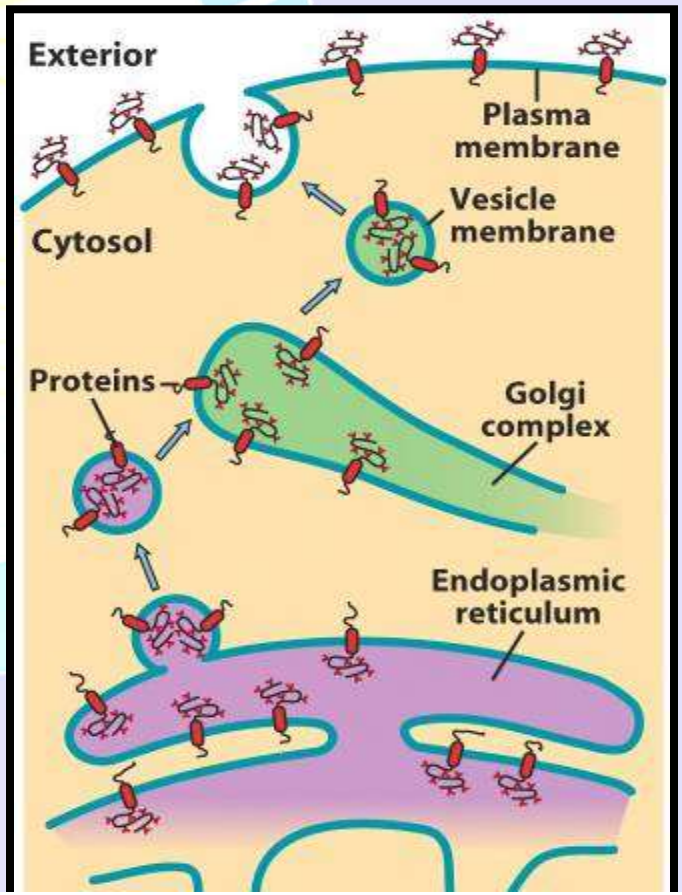




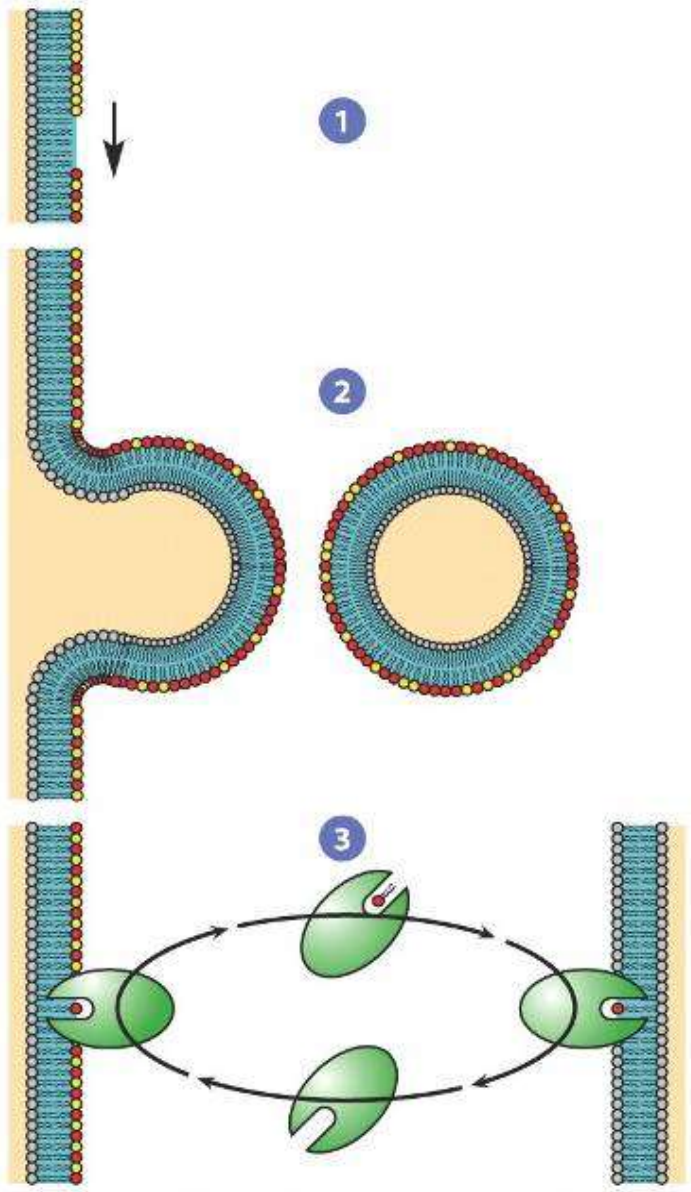
ROUGH ENDOPLASMIC RETICULUM



Phospholipid synthesis in the rough ER



ROUGH ENDOPLASMIC RETICULUM

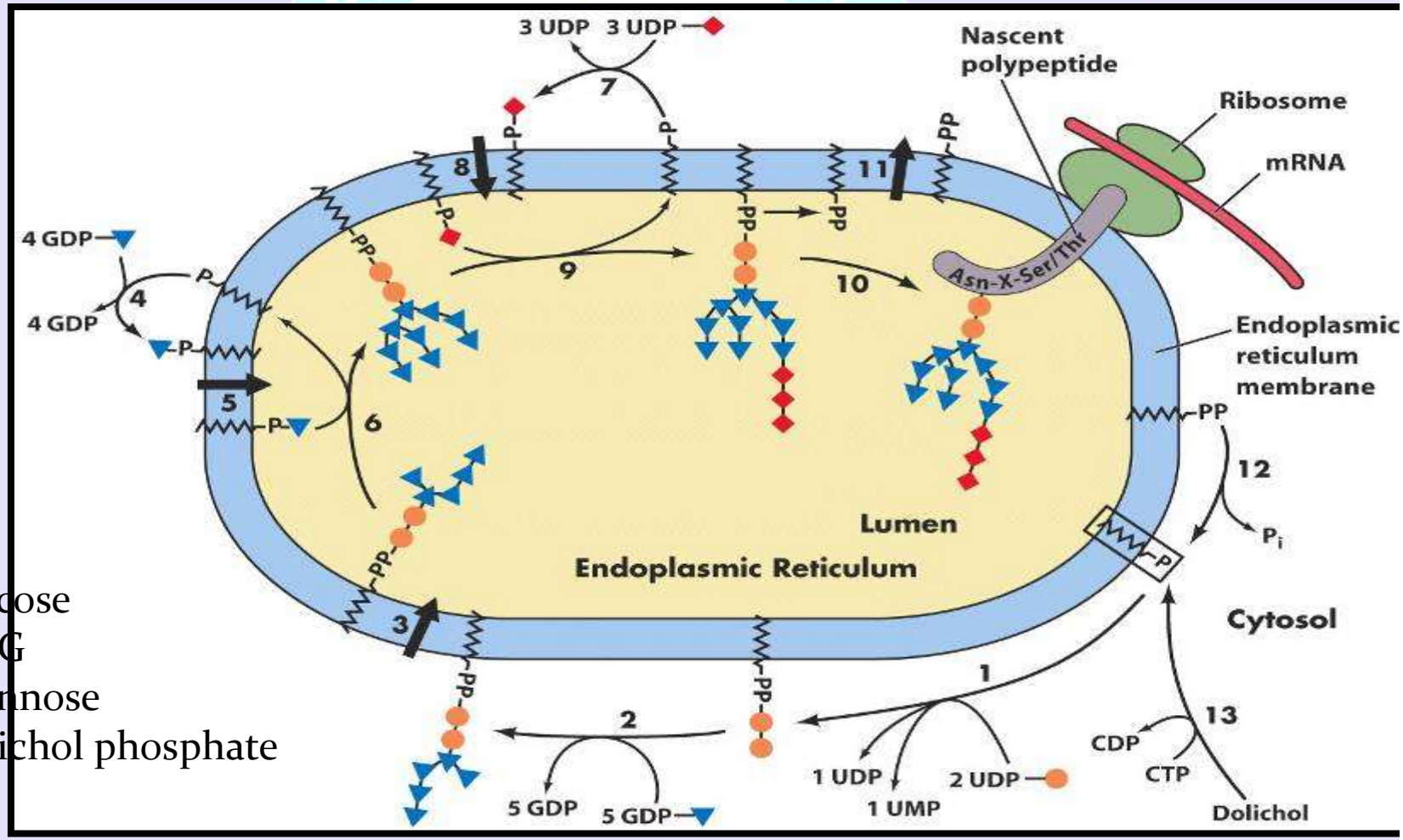


Phospholipid synthesis in the rough ER

1. Head groups are modified enzymatically
2. Vesicle has distinct phopholipid composition than ER membrane
3. Phospholipids removed from one membrane and inserted into another by phospholipid transferases)

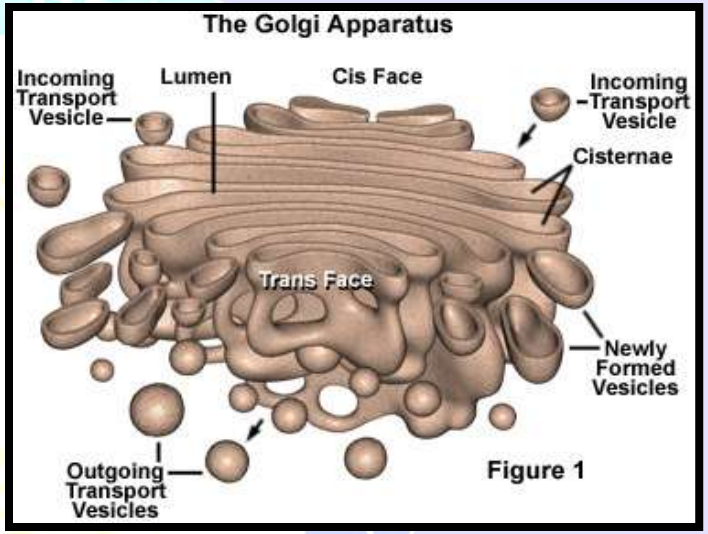
ROUGH ENDOPLASMIC RETICULUM

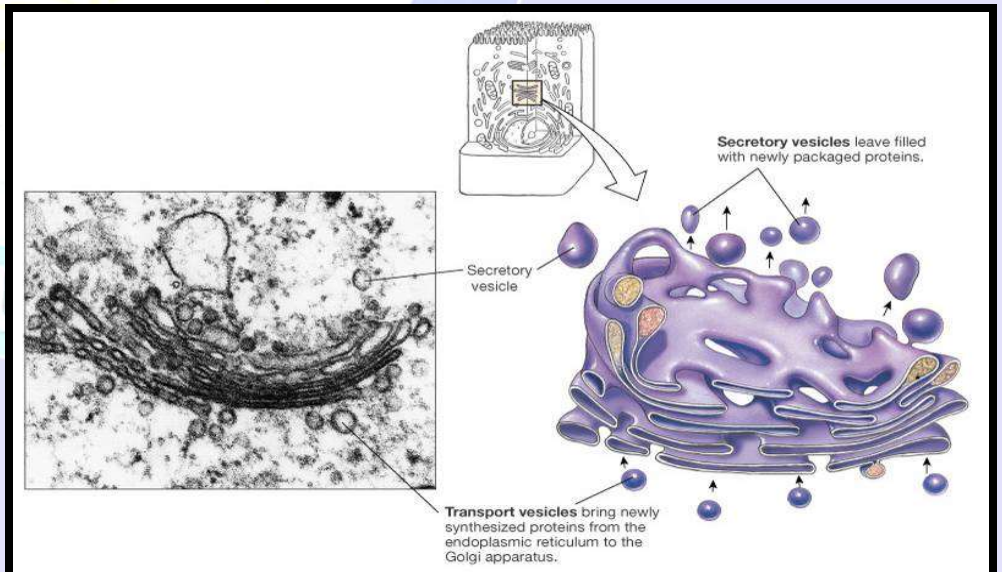
Carbohydrate synthesis in the rough ER



GOLGI APPARATUS

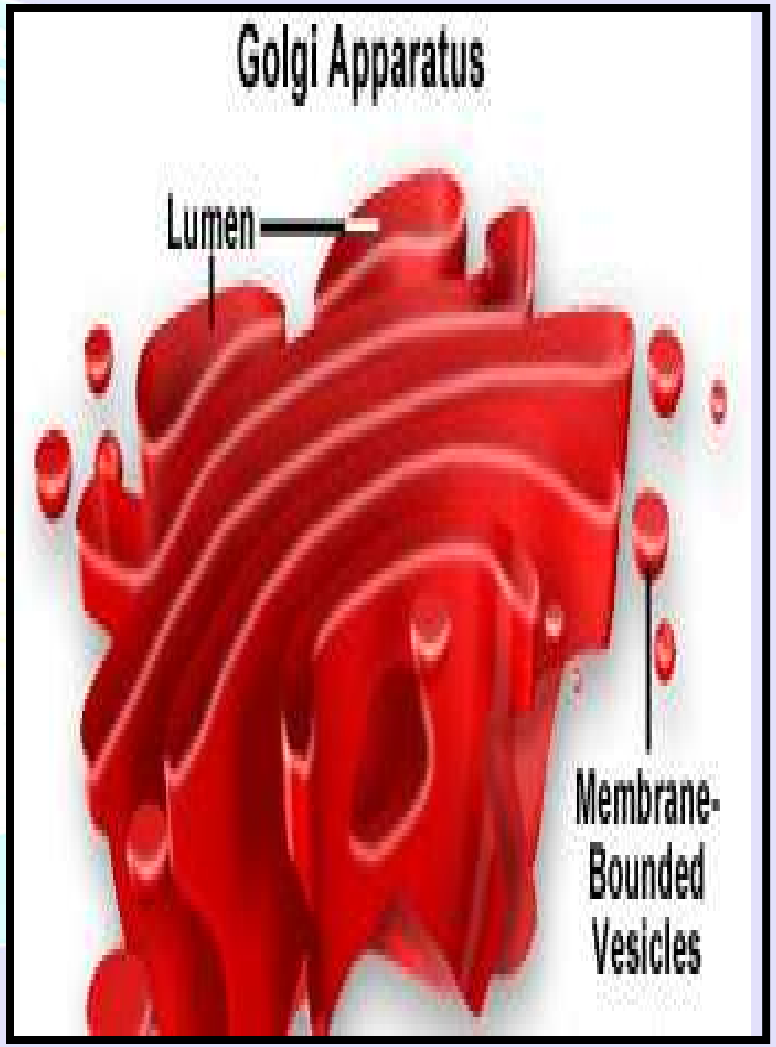
- The Golgi apparatus (also Golgi body or the Golgi Complex) is an organelle found in most eukaryotic cells





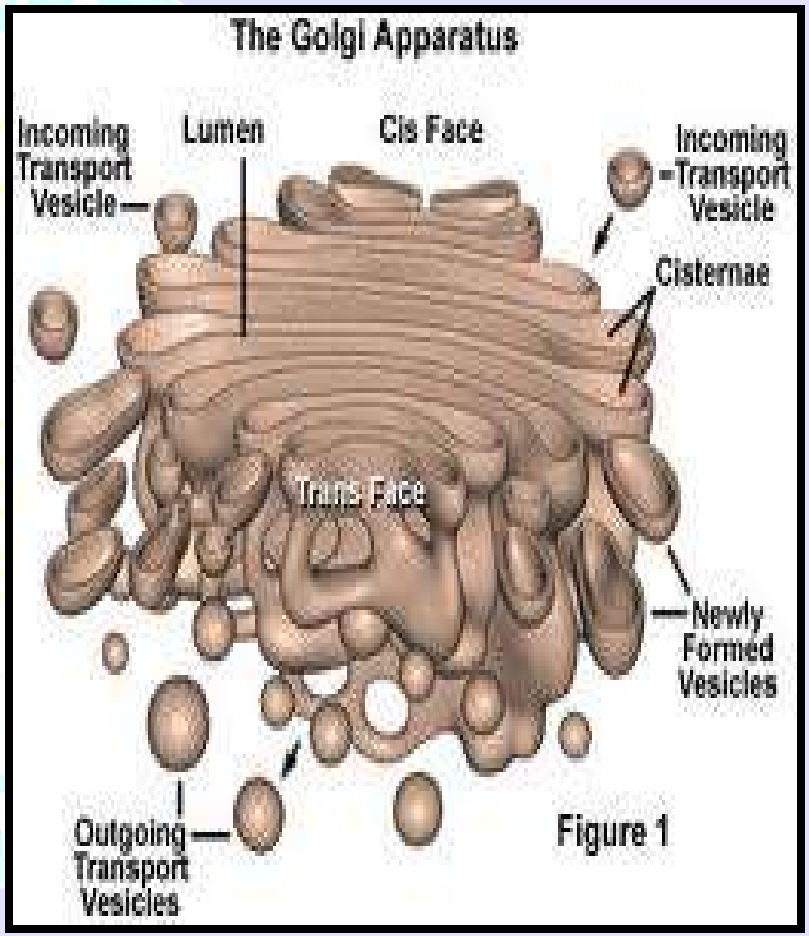
GOLGI APPARATUS

- The Golgi is composed of stacks of membrane-bound structures known as cisternae (singular: cisterna).
- An individual stack is sometimes called a dictyosome



FUNCTION OF THE GOLGI APPARATUS

- To process and package macromolecules, such as proteins and lipids,.
- After their synthesis and before they make their way to their destination; it is particularly important in the processing of proteins for secretion.
- forms a part of the cellular endomembrane system.

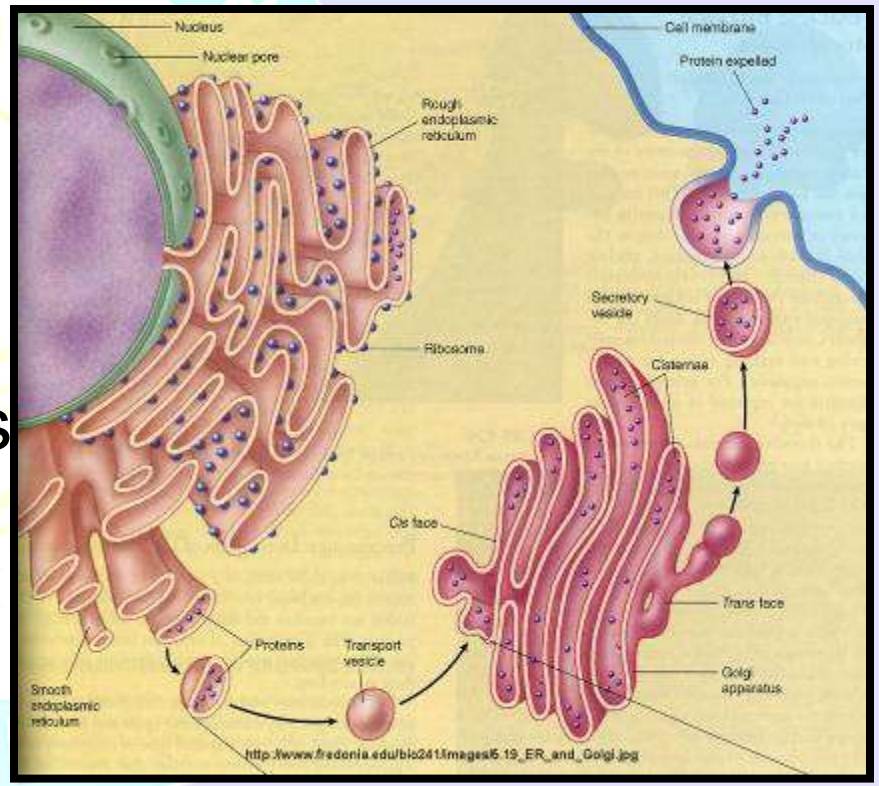


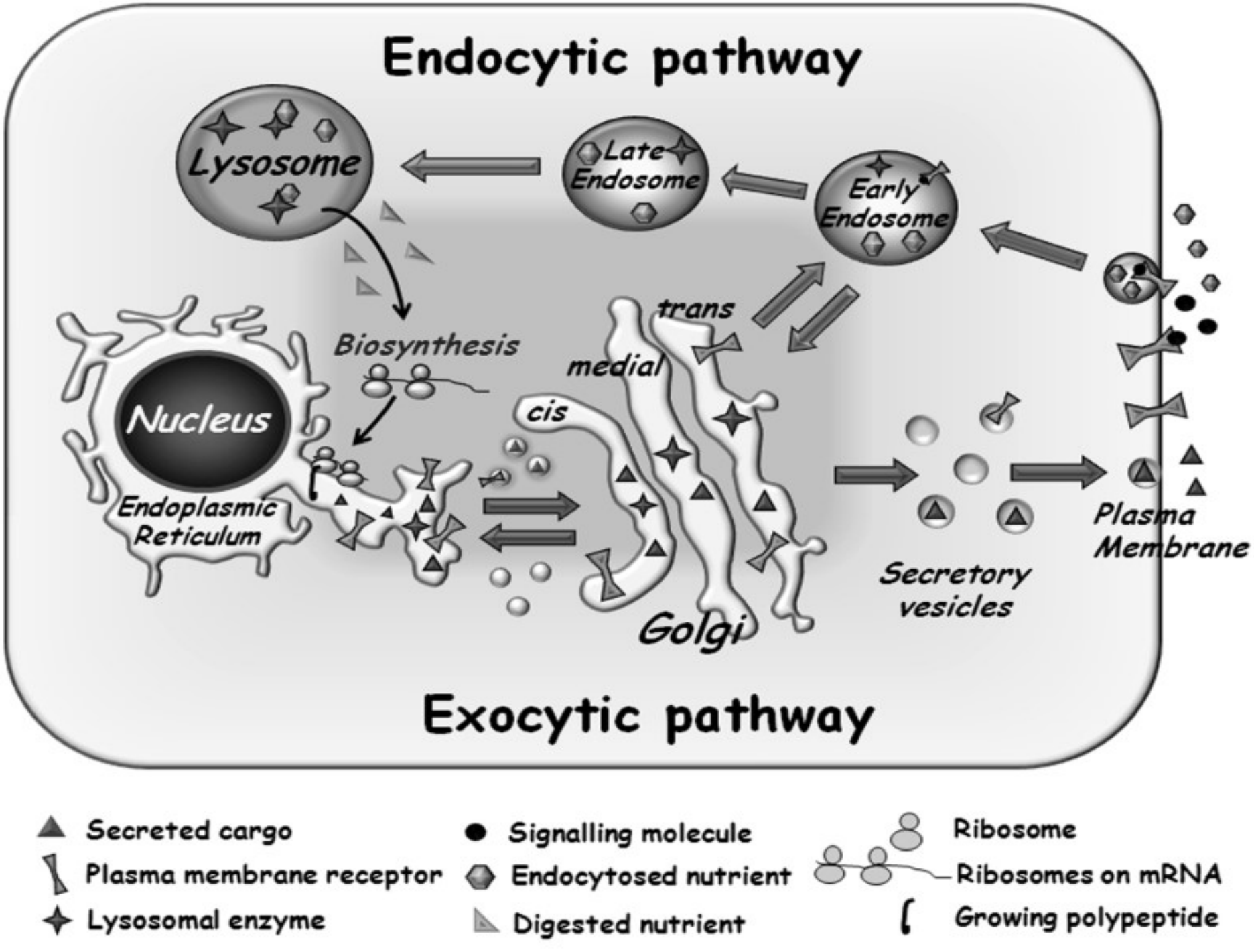
FUNCTION OF THE GOLGI APPARATUS

- Has a putative role in apoptosis
- Plays an important role in the synthesis of proteoglycans.
- Cells synthesize a large number of different macromolecules.
- Integral in modifying, sorting, and packaging these macromolecules for cell secretion (exocytosis) or use within the cell.

GOLGI APPARATUS DURING MITOSIS

- The Golgi apparatus will break up and disappear following the onset of mitosis, or cellular division.
- During the telophase of mitosis, the Golgi apparatus reappears.



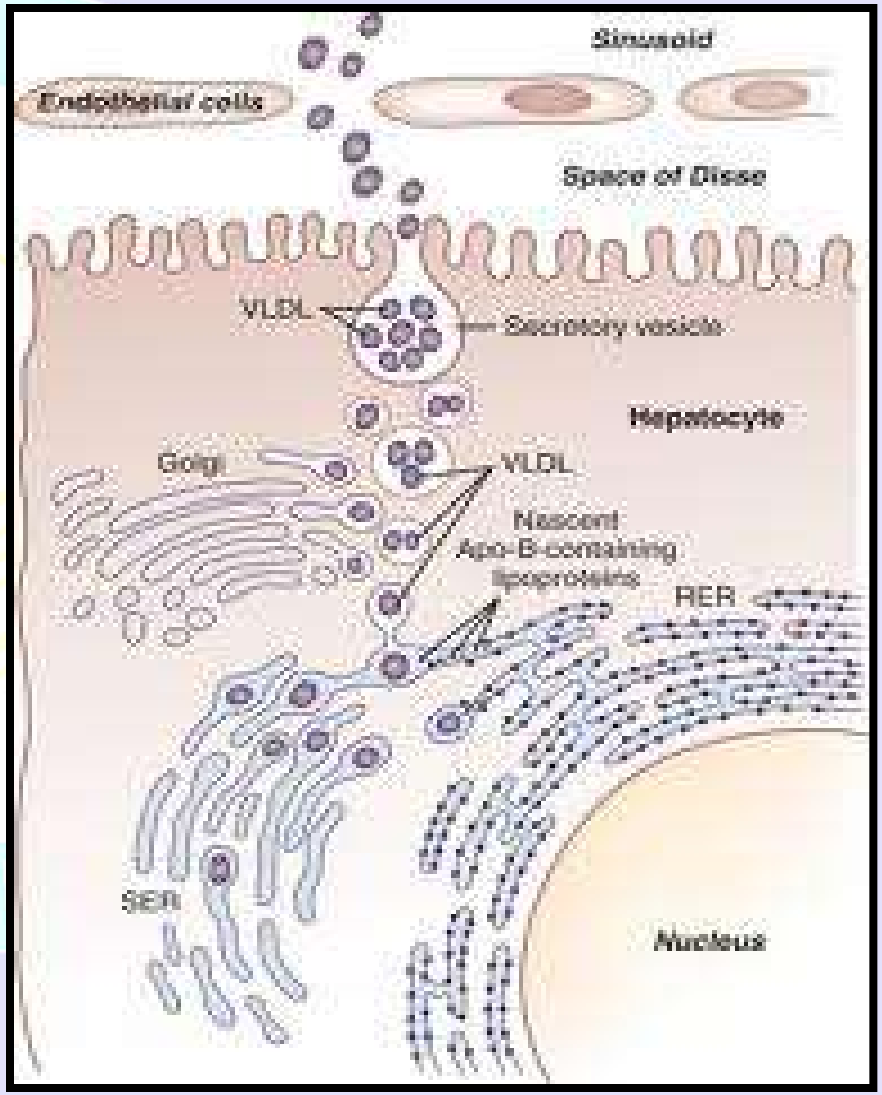

Endocytic pathway
Lysosome
Late Endosome
Early Endosome
Biosynthesis
trans
medial
cis
Nucleus
Endoplasmic Reticulum
Golgi
Secretory vesicles
Plasma Membrane
Exocytic pathway
Secreted cargo
Plasma membrane receptor
Lysosomal enzyme
Signalling molecule
Endocytosed nutrient
Digested nutrient
Ribosome
Ribosomes on mRNA
Growing polypeptide

EXOCYTIC VESICLES

- Vesicle contains proteins destined for extracellular release.
- After packaging the vesicles bud off and immediately move towards the plasma membrane.
- Where they fuse and release the contents into the extracellular space in a process known as constitutive secretion.
- Antibodies release by activated plasma B cells.

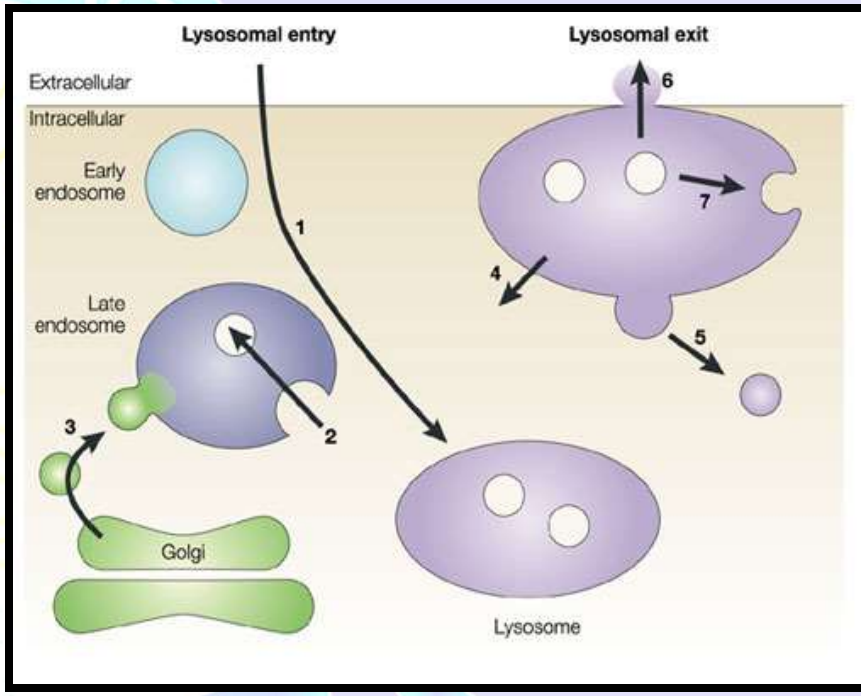
SECRETORY VESICLES

- Vesicle contains proteins destined for extracellular release.
- After packaging the vesicles bud off and are stored in the cell until a signal is given for their release.
- When the appropriate signal is received they move towards the membrane and fuse to release their contents.
- This process is known as regulated secretion.
- Nurotransmitter release from neurons.



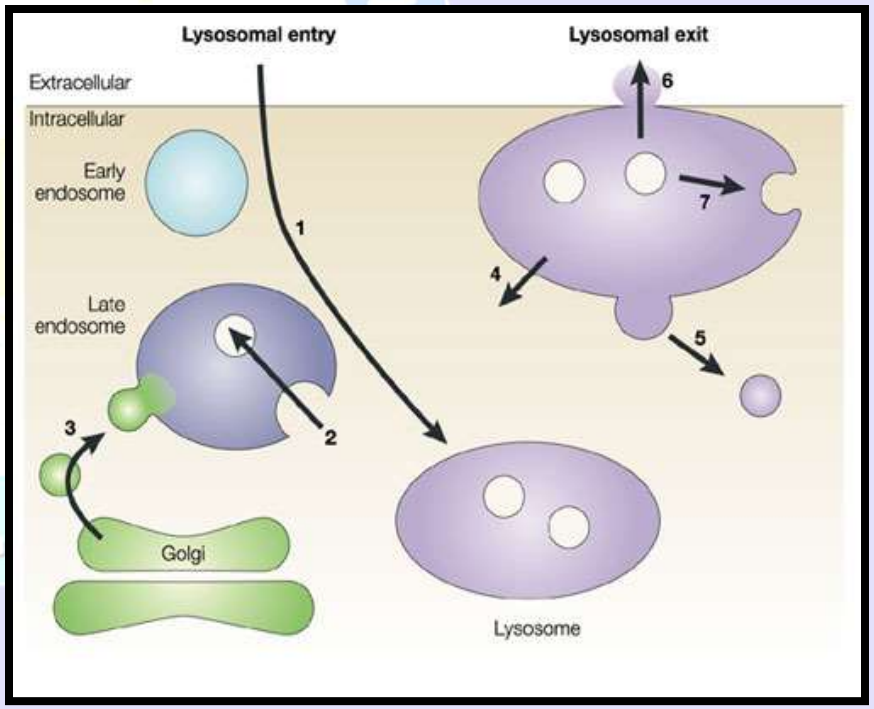
LYSOSOMAL VESICLES

- Vesicle contains proteins destined for the lysosome, an organelle of degradation containing many acid hydrolases, or to lysosome-like storage organelles.
- These proteins include both digestive enzymes and membrane proteins.
- The vesicle first fuses with the late endosome, and the contents are then transferred to the lysosome via unknown mechanisms.
- Digestive proteases destined for the lysosome.



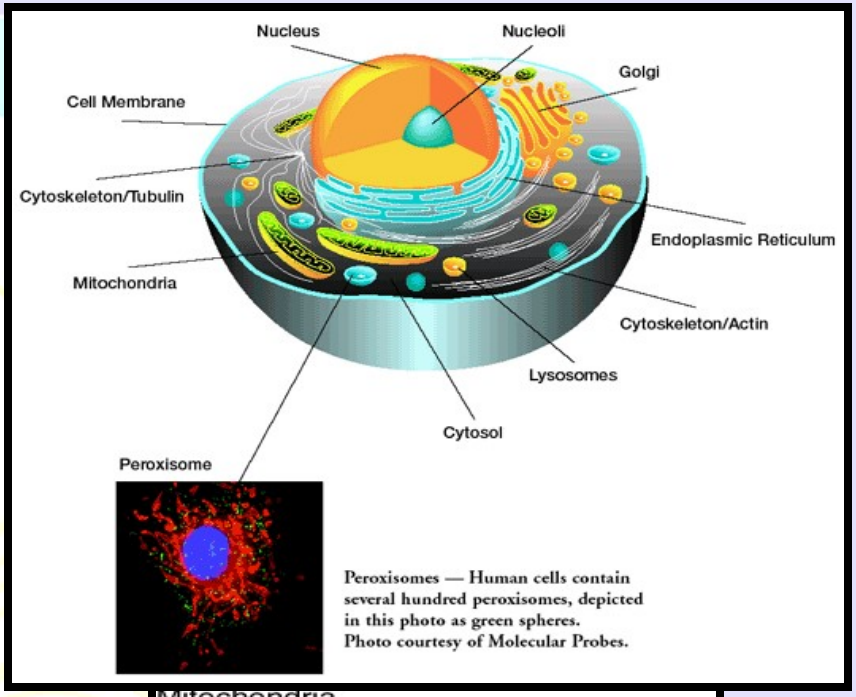
LYSOSOMAL VESICLES

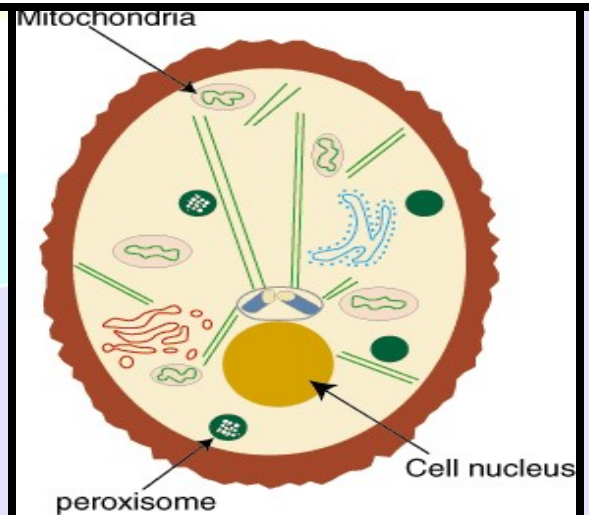
- The vesicle first fuses with the late endosome, and the contents are then transferred to the lysosome via unknown mechanisms.
- Digestive proteases destined for the lysosome.



PEROXISOMES

- Peroxisomes are vesicles that contain enzymes.
- The enzymes in these organelles use up oxygen and produce hydrogen peroxide.
- Peroxisomes are abundant in the liver where they produce bile salts and cholesterol and break down fats.

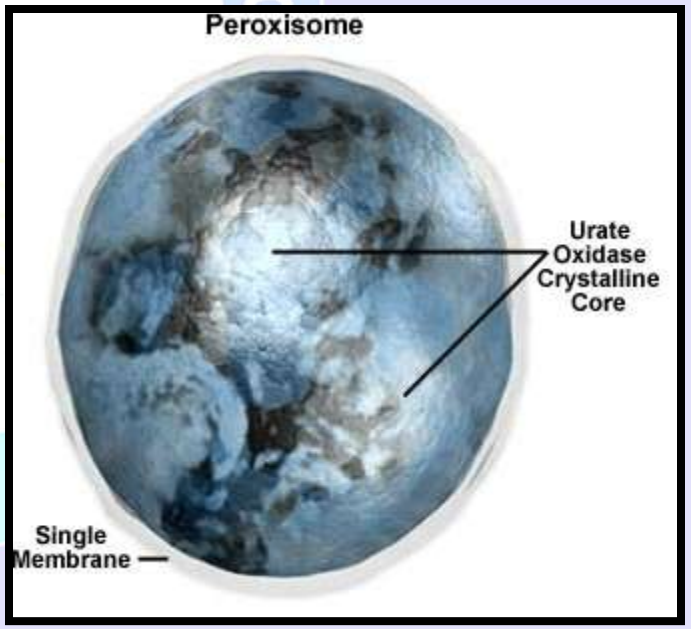




PEROXISOMES

- Enzymes oxidize toxins and remove free-radicals.

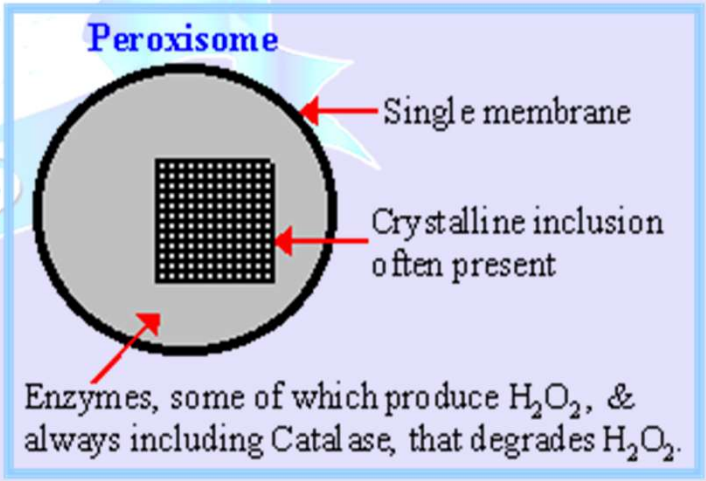
- membrane-walled sacs of oxidase enzymes
  - Enzymes neutralize **free radicals** and break down poisons
  - Break down long chains of fatty acids



PEROXISOME FUNCTION.

- A major function of the peroxisome is the breakdown of very long chain fatty acids through beta-oxidation.

- Peroxisomes contain oxidative enzymes, such as catalase, D-amino acid oxidase, and uric acid oxidase



CYTOPLASM

- The **cytoplasm** is a small gel-like substance residing between the cell membrane holding all the cell's internal sub-structures (called organelles), except for the nucleus.
- All the contents of the cells of prokaryote organisms (which lack a cell nucleus) are contained within the cytoplasm.
- Within the cells of eukaryote organisms the contents of the cell nucleus are separated from the cytoplasm, and are then called the nucleoplasm.

CYTOSOL

- The cytosol is the portion not within membrane-bound organelles.
- The cytosol is a translucent fluid in which the plasmic elements are suspended.
- Cytosol makes up about 70% of the cell volume and is composed of water, salts and organic molecules.
- The cytoplasm also contains the protein filaments that make up the cytoskeleton, as well as soluble proteins and small structures such as ribosomes, proteasomes.
- The inner, granular and more fluid portion of the cytoplasm is referred to as endoplasm.

REFERENCES

- *BASIC HISTOLOGY BY JUNQUEIRA*

10TH EDITION

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